A FLAT RING MAP OF FINITE PRESENTATION IS PERFECT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The counted statement span is a single sentence; the counted proof span opens with the author's sentence 'It is clear that B has finite tor dimension.' and that assessment is delivered verbatim and not expanded. The three references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Approximation of a flat finitely presented ring map by a flat finite type map of finite type Z-algebras. invoked at proof lines 21822-21825. Produces the Noetherian-flavoured model A_0 -> B_0 whose base change is A -> B.
- A finite type ring map of Noetherian rings, or of finite type Z-algebras, is relatively pseudo-coherent. invoked at proof lines 21825-21827. Gives pseudo-coherence of the approximating map.
- Base change of relative pseudo-coherence along tor independent ring maps. invoked at proof lines 21829-21831. Transfers pseudo-coherence from A_0 -> B_0 to A -> B.

Source reference. Original lemma 02JO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3343–3343. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 21715: sources/stacks-more-algebra/pinned-original-source.tex, source lines 21715–21715. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 21543: sources/stacks-more-algebra/pinned-original-source.tex, source lines 21543–21543. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *A ring map which is flat and of finite presentation is perfect.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $A \to B$ be a ring map which is flat and of finite presentation. It is clear that B has finite tor dimension. By Algebra, Lemma 02JO there exists a finite type $\mathbf{Z}$-algebra $A_0 \subset A$ and a flat finite type ring map $A_0 \to B_0$ such that $B = B_0 \otimes_{A_0} A$. By Lemma 21715 we see that $A_0 \to B_0$ is pseudo-coherent. As $A_0 \to B_0$ is flat we see that B_0 and A are tor independent over A_0, hence we may use Lemma 21543 to conclude that $A \to B$ is pseudo-coherent. $\square$